STAT 527 Advanced Regression Analysis
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Inference in Linear Regression

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Outline

- Determine conditions for applying t-tests and F-tests;
- The meaning of t-tests and F-tests, and their relationship;
- Execute t-tests and F-tests using R and interpret the results.

Inference

- $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ provide point estimates for the parameters $\beta_0, \beta_1, \ldots, \beta_p$.
- We do not know how for away the true parameters may be from the estimates.
- For example, if $\hat{\beta}_1 = 1.2$, how can we say with certain confidence β_1 cannot be equal to zero?
- Can we give a set of values containing $\hat{\beta}_1$ with certain confidence?

Inference

$\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_p$ provide point estimates for the parameters $\beta_0, \beta_1, \ldots, \beta_p$.

To draw conclusions about them (and their functions), we use Hypothesis testing and confidence intervals as inference tools.

We will learn about

- t-test for an individual model coefficient
- F-test for model reduction
- confidence interval for an individual model coefficient
- confidence interval for the mean of Y
- prediction interval for a new Y

Working Example

Savings rates in 50 countries

```
Description:

   The 'savings' data frame has 50 rows (countries) and 5 columns.
   The data is averaged over the period 1960-1970.

Usage:

      data(savings)
```

load dataset

```
Format:

     This data frame contains the following columns:

     'sr' savings rate - personal saving divided by disposable income

     'pop15' percent population under age of 15

     'pop75' percent population over age of 75

     'dpi' per-capita disposable income in dollars

     'ddpi' percent growth rate of dpi
```

Savings Data

Look at the first few lines of data ...

```
> library(faraway)

> head(savings)  # shows first 6 observations
             sr pop15 pop75     dpi ddpi
Australia 11.43 29.35  2.87 2329.68 2.87
Austria   12.07 23.32  4.41 1507.99 3.93
Belgium   13.17 23.80  4.43 2108.47 3.82
Bolivia    5.75 41.89  1.67  189.13 0.22
Brazil    12.88 42.19  0.83  728.47 4.56
Canada     8.79 31.72  2.85 2982.88 2.43
```

We will consider a model with savings rate (`sr`) as the dependent variable, and the other variables as independent variables.

That is, we would like to explain (mean) savings rate in terms of the other variables.

Use **library()** when you want to stop and get an error if the package is not available.
Use **require()** when you want to check if a package is available and potentially take a different action if it's not.

Full Model Fit

Fit the **full model** for a quick inspection ...

```
> fullmodel <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data=savings)

> summary(fullmodel)

...

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 28.5660865  7.3545161   3.884 0.000334 ***
pop15       -0.4611931  0.1446422  -3.189 0.002603 **
pop75       -1.6914977  1.0835989  -1.561 0.125530
dpi         -0.0003369  0.0009311  -0.362 0.719173
ddpi         0.4096949  0.1961971   2.088 0.042471 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.803 on 45 degrees of freedom
Multiple R-squared:  0.3385,Adjusted R-squared:  0.2797
F-statistic: 5.756 on 4 and 45 DF,  p-value: 0.0007904
```

Confidence Interval

A confidence interval for β_k with level $1 - \alpha$ is given by

$$(\hat{\beta}_k - t_{\alpha/2}\, se(\hat{\beta}_k),\ \ \hat{\beta}_k + t_{\alpha/2}\, se(\hat{\beta}_k))$$

where $t_{\alpha/2}$ refers to the $1 - \alpha/2$ percentile of a t-distribution with $n - p'$ degrees of freedom.

We can get these confidence intervals for the regression coefficients from R:

confint(object, param, level=0.95, ...)

```
> confint(fullmodel)
                    2.5 %       97.5 %
(Intercept) 13.753330728 43.378842354
pop15       -0.752517542 -0.169868752
pop75       -3.873977955  0.490982602
dpi         -0.002212248  0.001538444
ddpi         0.014533628  0.804856227
```

By default, these are 95% confidence intervals. (Specify the parameter `level` to get a different confidence level.)

Estimating Mean at a Future Observation

We are also often interested in estimating

$$E[Y \mid \boldsymbol{x}_*] = \boldsymbol{x}_*^T \boldsymbol{\beta},$$

and also obtain predictions for future observations at the predictor vector $\boldsymbol{x}_*$.

An unbiased estimator of $E[Y \mid \boldsymbol{x}_*]$ is

$$\hat{y}_* = \boldsymbol{x}_*^T \hat{\boldsymbol{\beta}} = \boldsymbol{x}_*^T (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y}$$

This is a linear transformation of $\boldsymbol{y}$, so we can easily derive the variance of $\hat{y}_*$, and find its standard error.

Uncertainity in Estimation

The standard error of the fitted value is

$$sefit(\hat{y}_*) \;=\; \hat{\sigma}\sqrt{\boldsymbol{x}_*^T(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{x}_*}$$

[Derive ...]

A confidence interval for $E[Y \mid \boldsymbol{x}_*]$ is

$$(\hat{y}_* - t_{\alpha/2}\, sefit(\hat{y}_*),\; \hat{y}_* + t_{\alpha/2}\, sefit(\hat{y}_*))$$

Prediction at a Future Observation

Also, $\hat{y}_*$ provides a *point prediction* for a *new* observation of Y at $\boldsymbol{x}_*$. In order to find a **prediction interval**, we need to consider *both* the variance in estimating $\hat{\boldsymbol{\beta}}$ *and* the variance associated with a new observation, which is σ^2.

The *standard error of prediction* is

$$sepred(\hat{y}_*) = \hat{\sigma}\sqrt{1 + \boldsymbol{x}_*^T(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{x}_*}.$$

We can obtain a prediction interval for a new observation of Y at $\boldsymbol{x}_*$ that has a $1-\alpha$ chance of including Y itself:

$$(\hat{y}_* - t_{\alpha/2}\, sepred(\hat{y}_*),\ \hat{y}_* + t_{\alpha/2}\, sepred(\hat{y}_*))$$

No matter how large the sample size is, width of a prediction interval, unlike a confidence interval, will not approach 0.

Analysis of the Prediction Interval

- We want to show that

$$\mathbb{P}[y_* \in [\hat{y}_* - t_{\alpha/2} sepred(\hat{y}_*),\ \hat{y}_* - t_{\alpha/2} sepred(\hat{y}_*)]] \geq 1-\alpha;$$

- Recall $y_* = x_*^\top \beta + \epsilon_*$;
- Equivalently, we want

$$\mathbb{P}[|y_* - \hat{y}_*| \leq t_{\alpha/2} sepred(\hat{y}_*)] \geq 1 - \alpha$$

but $y_* - \hat{y}_* = \epsilon_* + x_*^\top \beta - x_*^\top \hat{\beta}$.

Confidence Interval and Prediction Interval

Let's find a 95% confidence interval for the mean savings rate at the mean value of the predictors, and also compute a 95% prediction interval.

First, find the mean values (averages) of each of the predictors:

```
> head(savings)
             sr pop15 pop75     dpi ddpi
Australia 11.43 29.35  2.87 2329.68 2.87
Austria   12.07 23.32  4.41 1507.99 3.93
Belgium   13.17 23.80  4.43 2108.47 3.82
Bolivia    5.75 41.89  1.67  189.13 0.22
Brazil    12.88 42.19  0.83  728.47 4.56
Canada     8.79 31.72  2.85 2982.88 2.43

> apply(savings[,2:5], 2, mean)  # (column) means for variables 2 thru 5
   pop15     pop75       dpi      ddpi
 35.0896    2.2930 1106.7584    3.7576
```

submatrix

Margin =1: manipulation on rows
=2: on columns

apply function in R to data frame or matrix and gives vector output

Confidence Interval and Prediction Interval

Now find a confidence and a prediction interval at those (mean) values:

```
> # create a data frame where you would like prediction:

> x <- data.frame(pop15=35.0896, pop75=2.293, dpi=1106.758, ddpi=3.7576)

> # find a 95% (default level) confidence interval for mean:

> predict(fullmodel, x, interval="confidence")
    fit      lwr      upr
1 9.671 8.587859 10.75414

> # find a 95% prediction interval:

> predict(fullmodel, x, interval="prediction")
    fit      lwr      upr
1 9.671 1.935822 17.40618
```

Hypothesis Testing for a single coefficient

Full model:

$$sr = \beta_0 + \beta_{pop15} \times pop15 + \beta_{pop75} \times pop75 + \beta_{dpi} \times dpi + \beta_{ddpi} \times ddpi$$

- Null hypothesis (NH): $\beta_{pop75} = 0$
- Alternative hypothesis (AH): $\beta_{pop75} \neq 0$

Hypothesis Testing Recap

Typical procedure for hypothesis testing:

- Set $\alpha = P(\text{type I error})$. Typically $\alpha = 0.05$. α is called the significance level.
- Compute p-value: the probability of observed or more extreme departure from NH (in favor of AH) when NH is true. The departure from NH is usually measured using a test statistic.
- If p-value $< \alpha$, reject NH.

Z and t statistics

Test statistic for $NH: \quad \beta_j = 0$ vs $\quad AH: \beta_j \neq 0$

- if σ^2 known:

$$Z := \frac{\hat{\beta}_j}{sd(\hat{\beta}_j)},$$

if σ^2 not known:

$$t := \frac{\hat{\beta}_j}{se(\hat{\beta}_j)},$$

where

$$sd(\hat{\beta}_j) = \sqrt{(X^T X)^{-1}_{jj}\sigma^2}; \quad se(\hat{\beta}_j) = \sqrt{(X^T X)^{-1}_{jj}\hat{\sigma}^2}$$

.

The t-distribution

If ϵ_i are independent and normally distributed, then

$$Z \sim N(0,1); \quad t \sim t_{n-(p+1)}.$$

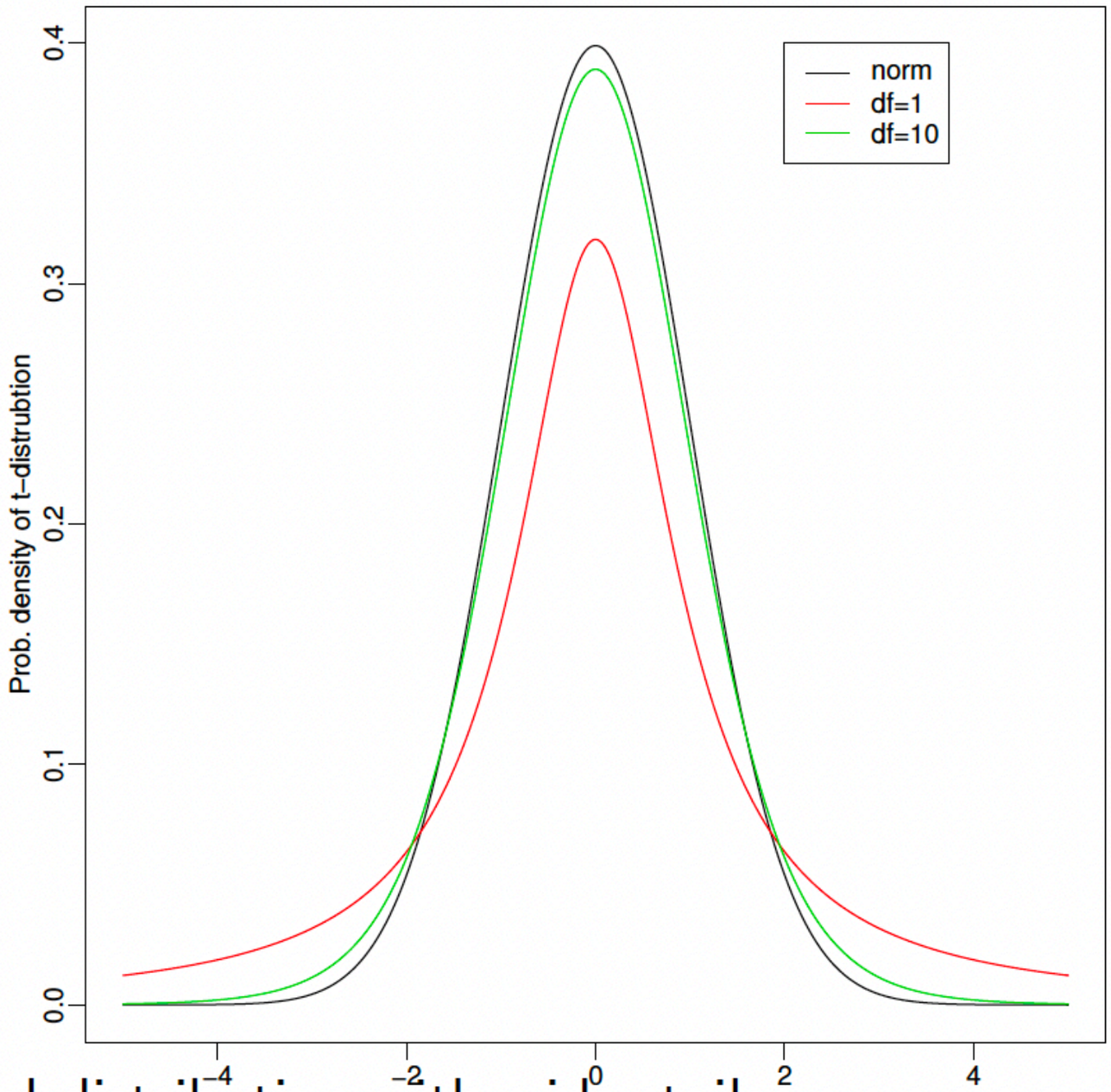
The t-distribution

Probability density function (pdf) of t_v:

$$t_v \quad : \quad \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{n\pi} \cdot \Gamma(\frac{v}{2})}(1 + z^2/v)^{-(v+1)/2}$$

- Has a single parameter v called **degrees of freedom**
- Symmetric, "bell-shaped", but heavier tails than normal
- As $v \to \infty$, $t_v \to N(0, 1)$
- $W = \frac{U}{\sqrt{\frac{V}{v}}}$, $W \sim t_v$ if $U \sim N(0, 1)$, $V \sim \chi^2_v$, and U, V are independent.

The t density



Like normal distribution with wider tails

The t-statistic (Savings Example)

- If the null is true, i.e., $\beta_{pop75} = 0$, then

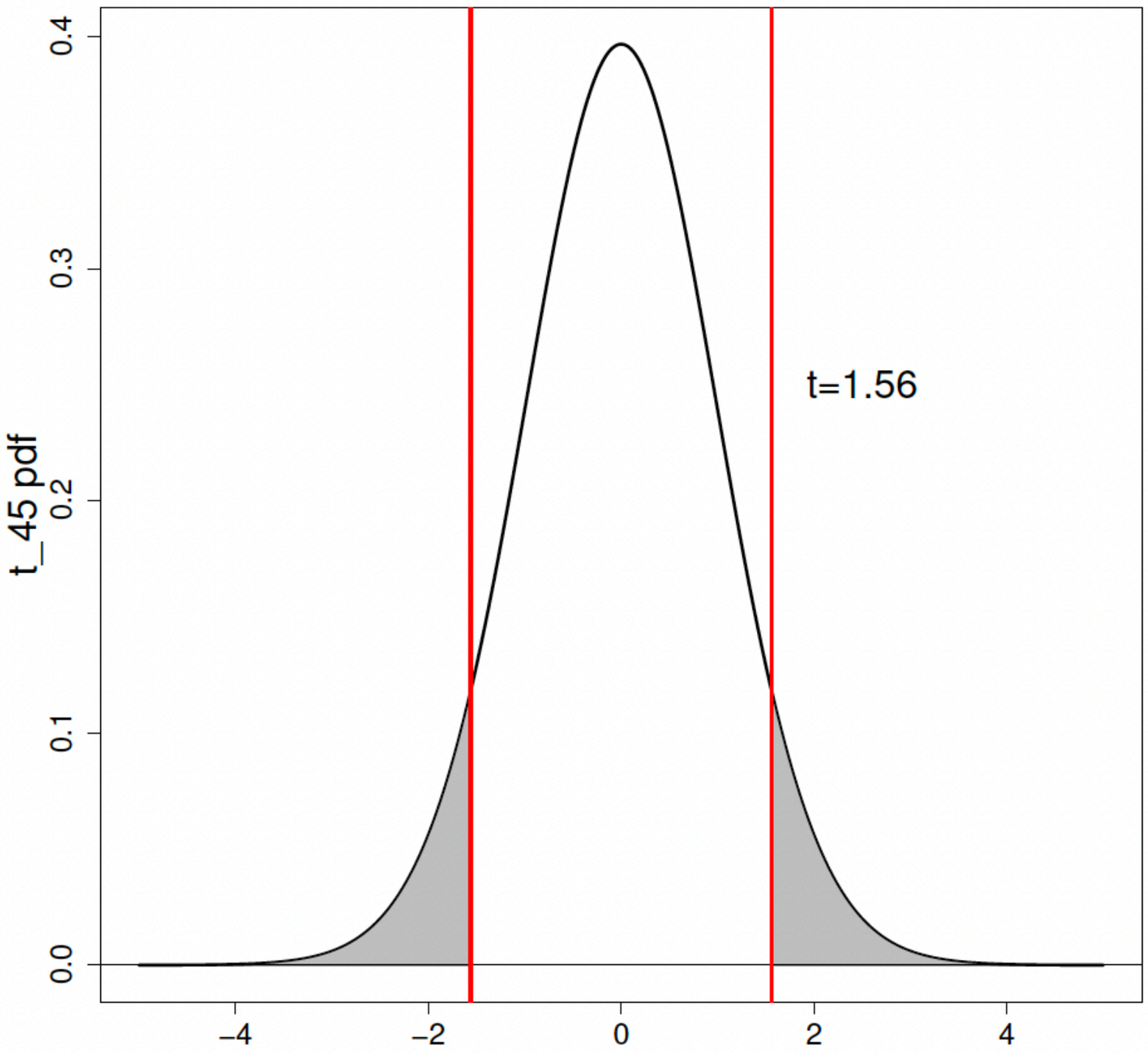
$$t = \frac{\hat{\beta}_{pop75}}{se(\hat{\beta}_{pop75})} \sim t_{50-(4+1)}$$

From the R output, we have $t = -1.56$

- P-value for two-sided test:
$P(|t_{45}| \geq 1.56) = 0.12553 > \alpha = 0.05$, therefore we fail to reject H_0.

- Thus *pop75* is not significant in the full model at level $\alpha = 0.05$.

t-statistic (Savings Example) Ctd



Reduced Model

Now suppose we want to test the possibility that both the age-related variables $pop15, pop75$ do not contribute to the model (and can be dropped). If they do not contribute, we can fit a **reduced model**:

```
> reducedmodel <- lm(sr ~ dpi + ddpi, data=savings)

> summary(reducedmodel)
...
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.3598478  1.2520610   5.080 6.43e-06 ***
dpi         0.0011949  0.0006091   1.962   0.0557 .
ddpi        0.5292495  0.2102883   2.517   0.0153 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.189 on 47 degrees of freedom
Multiple R-squared:  0.1616,Adjusted R-squared:  0.1259
F-statistic: 4.528 on 2 and 47 DF,  p-value: 0.01591
```

Comparing Models

How can we compare the results of the two fitted models?

More specifically, how would we test the following hypotheses?

```
Null Hypothesis:
     The age variables contribute nothing
     (the reduced model suffices)

Alternative Hypothesis:
     The age variables do contribute something
     (the full model is required)
```

Let's address this issue more formally and then return to the savings data.

Partitioning X and β

In matrix notation, partition $\boldsymbol{X} = (\boldsymbol{X}_1, \boldsymbol{X}_2)$, where $\boldsymbol{X}_1$ is $n \times (p' - q)$ and $\boldsymbol{X}_2$ is $n \times q$.

[Illustrate ...]

The corresponding partition of the regression parameter is $\boldsymbol{\beta}^T = (\boldsymbol{\beta}_1^T, \boldsymbol{\beta}_2^T)$, where $\boldsymbol{\beta}_1$ is $(p' - q) \times 1$ and $\boldsymbol{\beta}_2$ is $q \times 1$.

[Illustrate ...]

Testing for Reduced Model

Let the full model be written alternatively as

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \qquad \text{or} \qquad \boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{\beta}_1 + \boldsymbol{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$$

while the reduced model is

$$\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$$

These will be used for testing NH (null hypothesis) versus AH (alternative hypothesis) where

$$NH: \quad \boldsymbol{\beta}_2 = \mathbf{0}$$

and

$$AH: \quad \boldsymbol{\beta}_2 \neq \mathbf{0}$$

F-Test for Reduced Model

- F-test is used to test

$$NH : \boldsymbol{\beta}_2 = \mathbf{0} \text{ vs } AH : \boldsymbol{\beta}_2 \neq \mathbf{0},$$

.

- To perform F-test, fit the reduced model (under NH) and obtain its residual sum of squares RSS_{NH} with degrees of freedom $df_{NH} = n - (p' - q)$.

- Then, fit the full model (under AH) and obtain RSS_{AH} with $df_{AH} = n - p'$.

  The F-test Statistic is

$$F = \frac{(RSS_{NH} - RSS_{AH})/(df_{NH} - df_{AH})}{RSS_{AH}/df_{AH}}$$

F-Test for Reduced Model

- The F-test Statistic is

$$F = \frac{(RSS_{NH} - RSS_{AH})/(df_{NH} - df_{AH})}{RSS_{AH}/df_{AH}}$$

- Under NH, F statistic has an F distribution with q and $n - p'$ degrees of freedom under the assumption that the ε_i's are independent and $N(0, \sigma^2)$.

- Under AH, F will *tend to be larger*, so we reject NH when F is sufficiently large such that the type I error is α.

A Generalization

- The reduced model can assume other forms of linear constraints such as a set of coordinates are equal, instead of a set of coordinates are zero.

- Example: Null hypothesis $\mathsf{pop15} = \mathsf{pop75}$ (instead of $\mathsf{pop15} = \mathsf{pop75} = 0$).

- We then compute RSS_{NH} in a similar way and use the same formula for the F-statistic.

F-distribution

- $Z_1, \ldots, Z_k$ i.i.d. Normal(0,1). Then

$$U = Z_1^2 + \cdots + Z_k^2$$

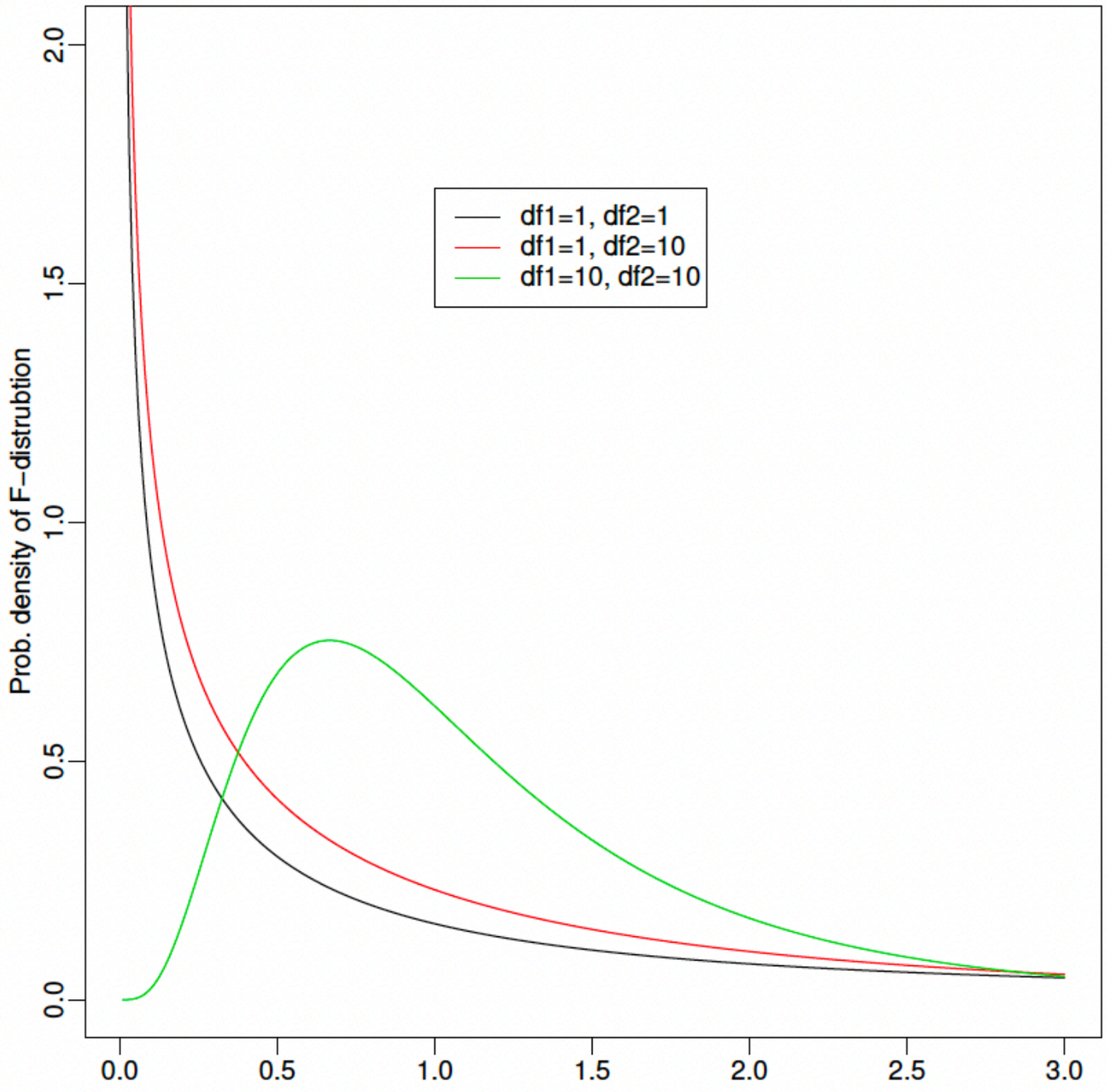
has χ^2 (chi-square) distribution with k degrees of freedom.

- χ^2_k is the same as $Gamma(k/2, 2)$.
- Suppose $U \sim \chi^2_k$, $W \sim \chi^2_m$ are independent. Then

$$\frac{U/k}{W/m} \sim F_{k,m}$$

F-distribution with k and m degrees of freedom.

F-distribution



Important facts: (1) $F_{k,m} > 0$ (2) $t_k^2 \sim F_{1,k}$

F Statistic in R

R has a function `anova` that computes F statistic and its p-value, based on the full and reduced models:

```
> anova(reducedmodel, fullmodel)
Analysis of Variance Table

Model 1: sr ~ dpi + ddpi
Model 2: sr ~ pop15 + pop75 + dpi + ddpi
  Res.Df    RSS Df Sum of Sq      F   Pr(>F)
1     47 824.72
2     45 650.71  2       174 6.0167 0.004835 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

[Conclusion ...]

(174/2) / (650.71/45)

F Statistic in Savings Data

In the savings data, consider the full model once again.

```
> summary(fullmodel)

...

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 28.5660865  7.3545161   3.884 0.000334 ***
pop15       -0.4611931  0.1446422  -3.189 0.002603 **
pop75       -1.6914977  1.0835989  -1.561 0.125530
dpi         -0.0003369  0.0009311  -0.362 0.719173
ddpi         0.4096949  0.1961971   2.088 0.042471 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.803 on 45 degrees of freedom
Multiple R-squared:  0.3385,Adjusted R-squared:  0.2797
F-statistic: 5.756 on 4 and 45 DF,  p-value: 0.0007904
```

Intercept-only Null Hypothesis

At the bottom of the output, there is an F-test for the null hypothesis that *none* of the variables contribute to the model, i.e. that the model with only an intercept will suffice.

So it is testing

$$NH: \quad \beta_{\texttt{pop15}} = \beta_{\texttt{pop75}} = \beta_{\texttt{dpi}} = \beta_{\texttt{ddpi}} = 0$$

versus

$AH:$ at least one of $\beta_{\texttt{pop15}}$, $\beta_{\texttt{pop75}}$, $\beta_{\texttt{dpi}}$, $\beta_{\texttt{ddpi}}$ is not zero

Q: What are the full and reduced models for this test?

Intercept-only reduced model

The reduced model for this test is

$$\texttt{sr} = \beta_0 + \varepsilon$$

and the full model is

$$\begin{aligned}\texttt{sr} = \beta_0 &+ \beta_{\texttt{pop15}} \cdot \texttt{pop15} + \beta_{\texttt{pop75}} \cdot \texttt{pop75} \\ &+ \beta_{\texttt{dpi}} \cdot \texttt{dpi} + \beta_{\texttt{ddpi}} \cdot \texttt{ddpi} + \varepsilon\end{aligned}$$

Relation between F-statistic and t-statistic

- Suppose we only wish to test that $NH : \beta_{\text{pop75}} = 0$.
- We could do an F-test as discussed now with 1 numerator degree of freedom or a t-test as before.
- The F-statistic is just the square of the t-statistic

$$t = \hat{\beta}_{\text{pop75}} / se(\hat{\beta}_{\text{pop75}}) = -1.56$$

- These statistics, are summarized in the R output table, with p-values obtained from referring to a t-distribution with 45 degrees of freedom.

Proof of the Equivalence

- We want to show that when $df_{NH} - df_{AH} = 1$, we have $F = t^2$.

- Then

$$F = \frac{RSS_{NH} - RSS_{AH}}{RSS_{AH}} df_{AH} \tag{1}$$

$$= \frac{\|Proj_{null(X_{\setminus j})/null(X)}(y)\|^2}{\|Proj_{null(X)}(y)\|^2} df_{AH} \tag{2}$$

- We can verify (by definitions) that a unit vector in the one-dimensional projection is $b := \frac{[X(X^\top X)^{-1}]_{:j}}{\sqrt{[(X^\top X)^{-1}]_{jj}}}$. Then

$$F = \frac{(b^\top y)^2}{\hat{\sigma}^2} \tag{3}$$

$$= \frac{[(X^\top X)^{-1} X^\top y]_j^2}{[(X^\top X)^{-1}]_{jj} \hat{\sigma}^2} \tag{4}$$

Relationship between F statistic and R^2

The F-statistic for the null hypothesis

$$\beta_1 = \beta_2 = \cdots = \beta_p = 0$$

is

$$F = \frac{\|\hat{Y} - \bar{Y}\|^2}{\|Y - \hat{Y}\|^2} \frac{n-p-1}{p} = \frac{R^2}{1-R^2} \frac{n-p-1}{p},$$

an increasing function of R^2.

Thank you!

We gratefully acknowledge the contributions of previous course instructors and various online sources for some of the images and slides presented.